ElasticDino: Dense Semantic Features using Elastic Deformation - Supplementary Material

A. Implementation Details

We train our model using batches drawn alternately from two datasets: Hypersim [13], which provides ground truth for intrinsic decomposition and normal estimation, and ImageNet-22k [6] for increased image diversity.

To avoid introducing additional datasets for semantic segmentation and depth prediction, we utilize pseudo-labels generated by pre-trained models, specifically, Mask2Former (COCO,ViT-Large) [2] and DepthAnything [17] respectively. These labels are computed for images across both of the training datasets.

To minimize parameter count and overhead on top of the base ViT prediction, we use depth-wise separable convolutions [3]. We observe no performance hit compared to regular convolutions, with significantly faster training and inference, and vastly reduced memory usage.

We train in mixed-precision using an 8-bit version of the AdamW optimizer [11], with parameters $\epsilon = 10^{-5}$ for stability, an effective batch size of 16 and an exponential learning rate decay starting at 10^{-3}. The final model was trained on a single A100 GPU for 24 hours, amounting to 3.42kg of CO2-equivalent emissions, fully offset by the cloud provider (Google Cloud Services), and computed using the MachineLearning Impact calculator presented in [9].

B. Further Quantitative Results

B.1. Open Vocabulary Evaluation

We evaluate our model on zero-Shot open-vocabulary segmentation following the method presented in MaskCLIP [18]. This training-free approach computes pixel-wise predictions by taking the argmax over similarity scores between visual features and text embeddings generated from dataset labels in textual templates. Following JAFAR [5], we evaluate on CLIP-ViT-B/16 and observe competitive results, demonstrating that **ElasticDino** preserves the original backbone features through its upsampling. Results against competitors that are compatible with CLIP-ViT-B/16 (JAFAR [5], AnyUp [16], LoftUp [7]) are reported in Table 1.

Table 1. Zero-Shot Open-Vocabulary segmentation on VOC and Ade20k datasets following the methodology used in JAFAR [5] and introduced in MaskCLIP [18] paper. Competitive results indicating good alignment with original internal CLIP features.

Method	VOC		ADE20K	
	mIoU(↑)	Acc(↑)	mIoU(↑)	Acc(↑)
Bilinear	51.25	68.18	9.05	25.15
JAFAR	**63.99**	**77.91**	<u>10.69</u>	28.06
AnyUp	57.59	73.40	10.71	**28.95**
LoftUp	38.73	55.29	7.64	32.16
ElasticDino (ours)	<u>58.43</u>	<u>73.89</u>	**10.75**	<u>28.28</u>

B.2. Positional Information Preservation

While positional encoding related artifacts are alleviated, positional information still persists and can be extracted more easily from the features compared to competing upscaling methods. We train a simple pixel-wise head to predict normalized (x, y) coordinates from upsampled features. Bilinear upsampling performs best, as expected for a linear target, but **ElasticDino** indeed preserves positional information. See accuracy in $[10^{-3}$ MSE] results below.

	JAFAR	FeatUp	AnyUp	LoftUp	Bilinear	Ours
$10^{-3}MSE$	5.3	11.5	4.2	64.0	3.2	4.5

B.3. Semantic Segmentation and Depth Estimation over CLIP

Following the comparison against other task agnostic upsampling methods (JAFAR [5], AnyUp [16], LoftUp [7], UPLift [15]) over DINOv2 [12] and DINOv3 [14], we also compare recent works over CLIP [18] and get competitive results (Table 2). We follow established evaluation protocols, as described by JAFAR [5], measuring mean Intersection over Union (mIoU) and pixel-level accuracy (Acc) for the semantic segmentation task over four datasets, and Root Mean Square Error (RMSE) and δ_1 (measuring the percentage of pixels with a relative error under 25% of the ground-truth) for the depth estimation task over one dataset.

Table 2. Comparison of models over CLIP backbone. Benchmarks are run using the method and code used for the JAFAR [5] paper. Bold = best, underline = second best.

CLIP	Semantic Segmentation								Depth Est.	
	COCO		VOC		ADE20K		Cityscapes		COCO	
	mIoU(↑)	Acc(↑)	mIoU(↑)	Acc(↑)	mIoU(↑)	Acc(↑)	mIoU(↑)	Acc(↑)	δ_1(↑)	RMSE(↓)
JAFAR [5]	**58.16**	**78.52**	<u>79.07</u>	**94.69**	**36.65**	**71.39**	**50.98**	**91.23**	<u>45.75</u>	<u>0.92</u>
AnyUp [16]	53.05	74.54	69.85	91.46	31.16	67.35	39.63	86.62	44.81	0.96
ED (ours)	<u>57.86</u>	<u>78.23</u>	**79.26**	<u>94.67</u>	<u>35.5</u>	<u>71.22</u>	<u>46.21</u>	<u>90.67</u>	**48.20**	**0.86**

B.4. Semantic Segmentation and Depth Estimation for no-seg and no-depth ElasticDino Variants

We evaluated **ElasticDino** variants trained without the depth and the segmentation tasks in the same framework used to evaluate the full variant against task agnostic competitors over DINOv2 [12]. This is to strengthen the claim that **ElasticDino** only uses these tasks for structural guidance and does not rely on task specific information. Performance remains strong on these held-out tasks (Table 3).

C. Further Qualitative Results

In this section we offer additional qualitative results.

Dedicated Segmentation models. We evaluate **ElasticDino** against two leading dedicated segmentation specific approaches Mask2Former [2] and OneFormer [8] over the Cityscapes[4] dataset. We showcase that even though our approach is task agnostic we surpass leading dedicated segmentation models using only our lightweight operations over the DINOv2[12] backbone, see Figure 2.

Training heads. **ElasticDino** employs different tasks to train the upscaler, by co-training a set of local and minimalistic 1×1 convolution heads, each performing a different dense task, considering pixels individually. In Figure 8, we demonstrate the results of these seven heads, both in their final step, where the features are of 256×256 resolution, and in an intermediate point in the network, where the resolution is upscaled from 32×32 to only 64×64. We further demonstrate qualitative results for intrinsic decomposition using the probe heads over the validation set of HyperSim [13] in Figure 1.

PCA feature visualization. In the feature maps comparison figures (Figures 6 and 7), we visualize the upscaled feature maps of **ElasticDino** using the first three components of a PCA decomposition and those of JAFAR [5], AnyUp [16], LoftUp [7], UPLift [15]. As can be seen, our method achieves crisp features and cleaner maps from different artifacts as opposed to the competitors. It can be noticed that **ElasticDino** is better at removing artifacts originating from spatial-encodings and separates far and near objects (the bars of the forklift) as well as appearance from geometry (the shield).

Visualization of deformations along upscaling process. Figures 4, 5 visualize the cumulative deformations through the process of feature upscaling for different images. Deformations are visualized using color wheel optical flow [1] and features are visualized through the first three PCA components. It can be seen that deformations help to make the edges more crisp and reduce artifacts in the feature maps.

Linear segmentation and depth probe heads. Finally, we also compare **ElasticDino** against JAFAR [5], AnyUp [16], LoftUp [7] in the framework suggested by JAFAR where linear probe heads are trained over the upsamplers for segmentation and depth tasks displaying the information encoded inside the upsampled features produced by the model allows achieving top results for dense tasks such as semantic segmentation (Figures 10, 12, 9, 11) and depth estimation (Figure 13). It can be seen that **ElasticDino** achieves superior results with cleaner edges and often detects details other models are not able to.

D. Experiments to understand degraded performance over Cityscapes[4]

We investigated our change in relative performance on Cityscapes [4], in contrast to all other datasets. We hypothesize it is because Cityscapes consists mainly of smaller objects and segments, connecting to our known limitation of filling "holes" in feature maps. To further investigate this we evaluated AP (Average Precision) metric for the different models over segments of different size categories Small (area $< 32^2$), Medium ($32^2 <$ area $< 96^2$), Large (area $> 96^2$) as taken from COCO [10] instance segmentation evaluation methodology (see Figure 3). We also mea-

Table 3. Comparison of no-seg, no-depth **ElasticDino** variants against fully trained model. Benchmarks are run using the method and code used for the JAFAR[5] paper. Bold = best, underline = second best.

DINOv2	**Semantic Segmentation**								**Depth Est.**	
	COCO		VOC		ADE20K		Cityscapes		COCO	
Method	mIoU↑	Acc↑	mIoU↑	Acc↑	mIoU↑	Acc↑	mIoU↑	Acc↑	δ_1 ↑	RMSE↓
ED (no-depth variant)	61.7	81.1	84.1	96.3	41.3	75.5	<u>60.1</u>	93.1	60.5	0.64
ED (no-seg variant)	<u>62.5</u>	<u>81.6</u>	<u>85.9</u>	**96.8**	<u>41.5</u>	<u>75.8</u>	59.8	<u>93.3</u>	**64.8**	**0.56**
ED (full variant)	**62.72**	**81.74**	**86.00**	<u>96.78</u>	**42.10**	**76.00**	61.33	93.44	<u>64.14</u>	<u>0.57</u>

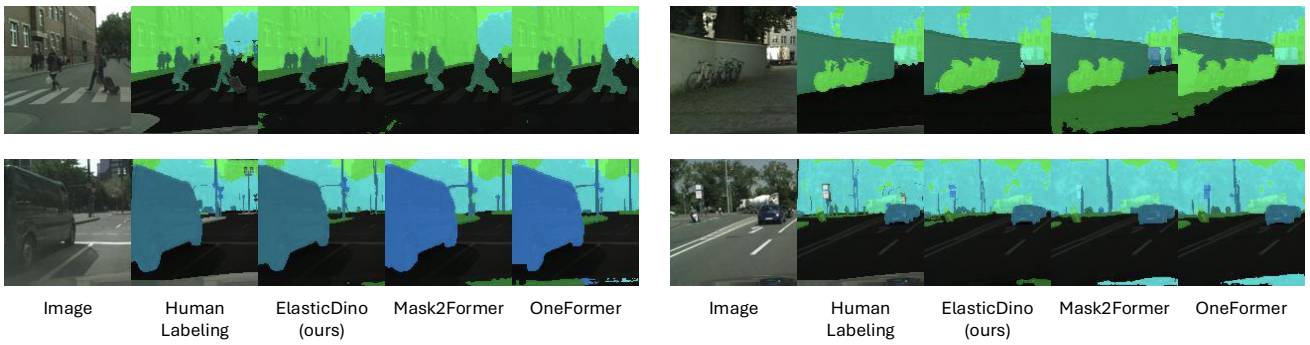


Figure 1. Qualitative comparison of **ElasticDino** to dedicated segmentation approaches (Mask2Former [2],OneFormer [8]) over the cityScapes dataset.

sured distribution of different segment size categories over the COCO[10], Cityscapes[4], Ade20k datasets. From this can see that the lower performance of **ElasticDino** can over Cityscapes can be explained by its lower performance on small thin objects present in this dataset in particular.

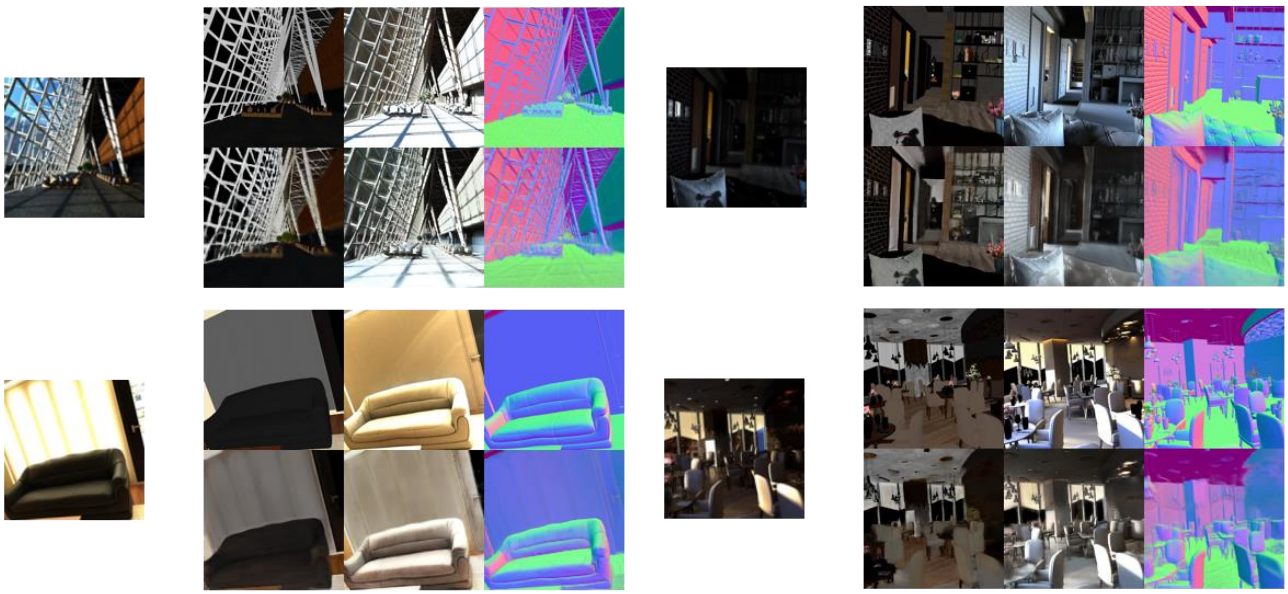


Figure 2. Intrinsic decomposition estimation trained on Hypersim [13] with a convolutional head at resolution 128x128. For each image, the decomposition is shown on the right: top row is the ground truth and bottom row is the prediction from ElasticDino. Left to right: albedo, diffuse shading, surface normals. Even in poor light conditions, we obtain strong qualitative results.

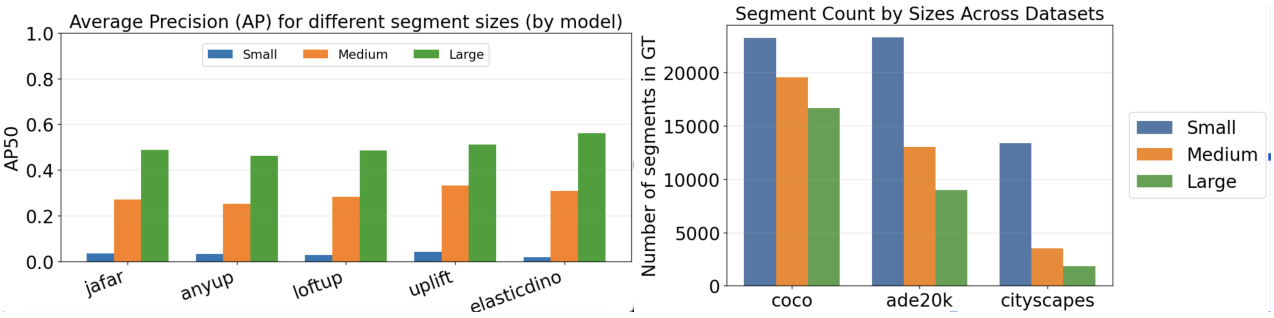
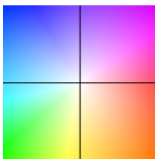


Figure 3. Left: Accuracy vs size. We perform better for larger segments. Right: sizes across datasets. Cityscapes consists of smaller objects.



Flow field color coding

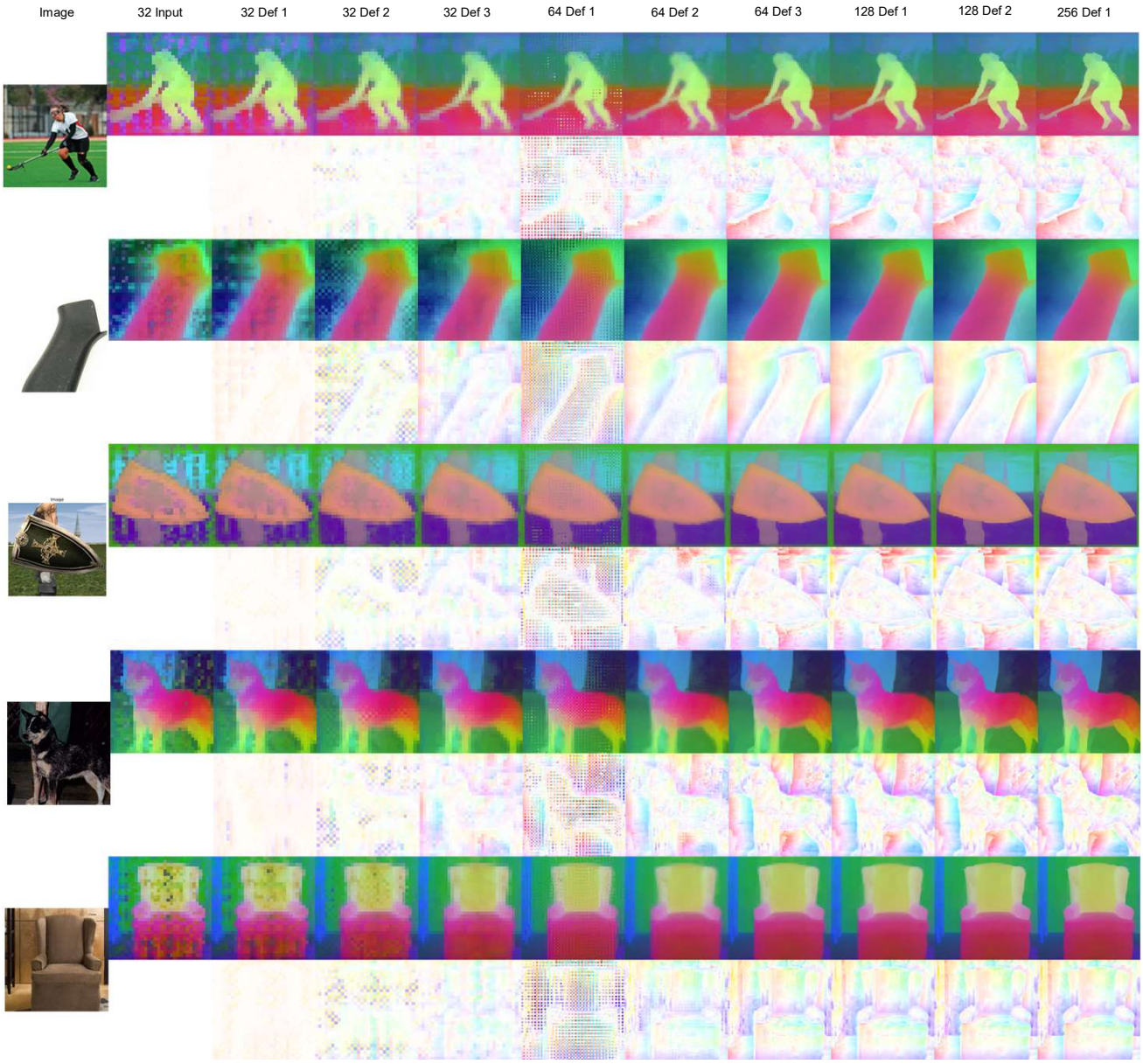
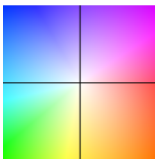


Figure 4. Visualization of deformations along the different resolution stages of the upscaling process and their effect on the features while being upscaled. Deformations are visualized using color wheel optical flow [1], showing where each pixel was sampled from. Features are visualized through the first three PCA components. For each image, top row shows the feature maps and bottom row shows the cumulative deformations so far. Zoomed-in viewing recommended.



Flow field color coding

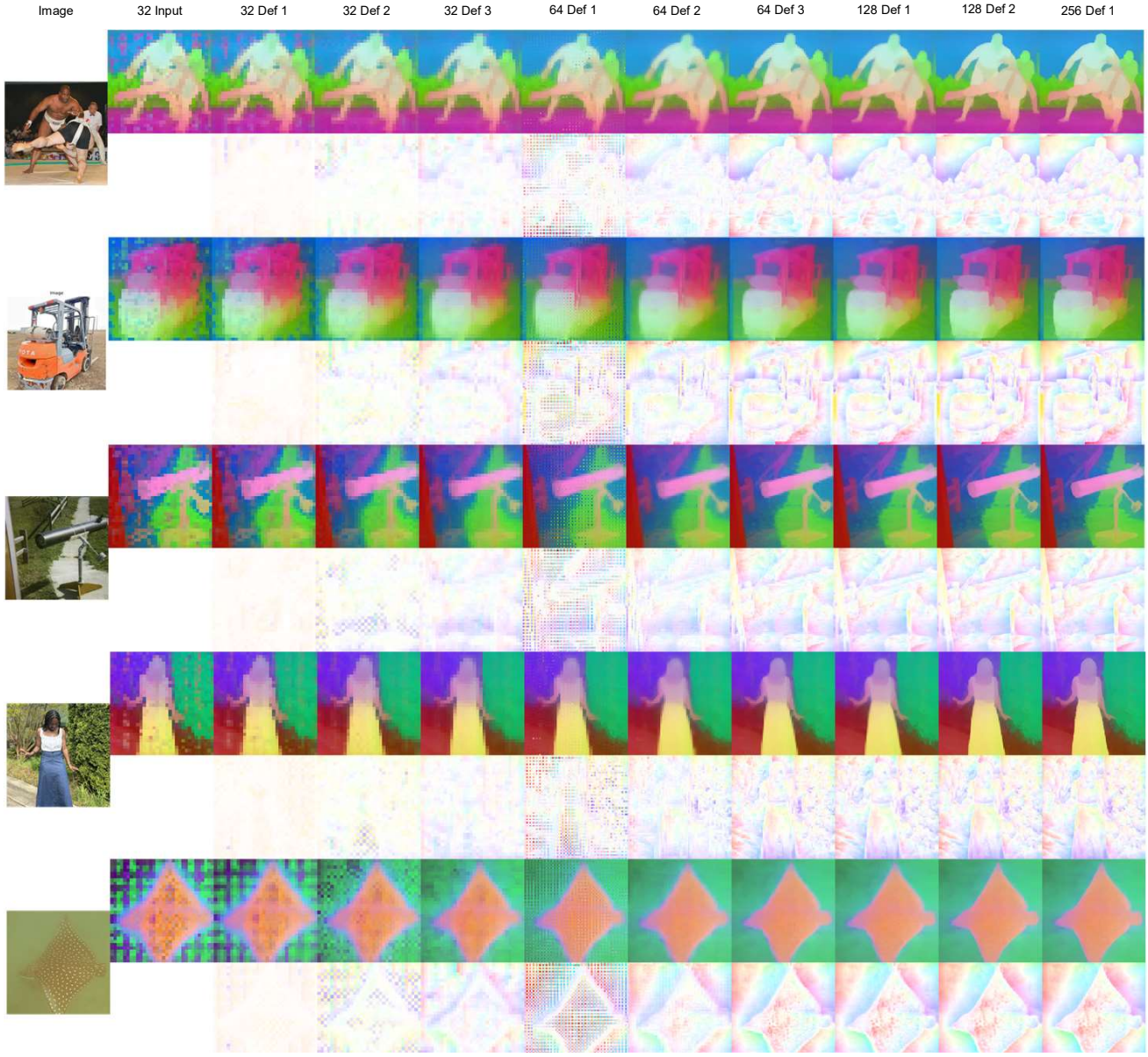


Figure 5. Visualization of deformations along the different resolution stages of the upscaling process and their effect on the features while being upscaled. Deformations are visualized using color wheel optical flow [1], showing where each pixel was sampled from. Features are visualized through the first three PCA components. For each image, top row shows the feature maps and bottom row shows the cumulative deformations so far. Zoomed-in viewing recommended.

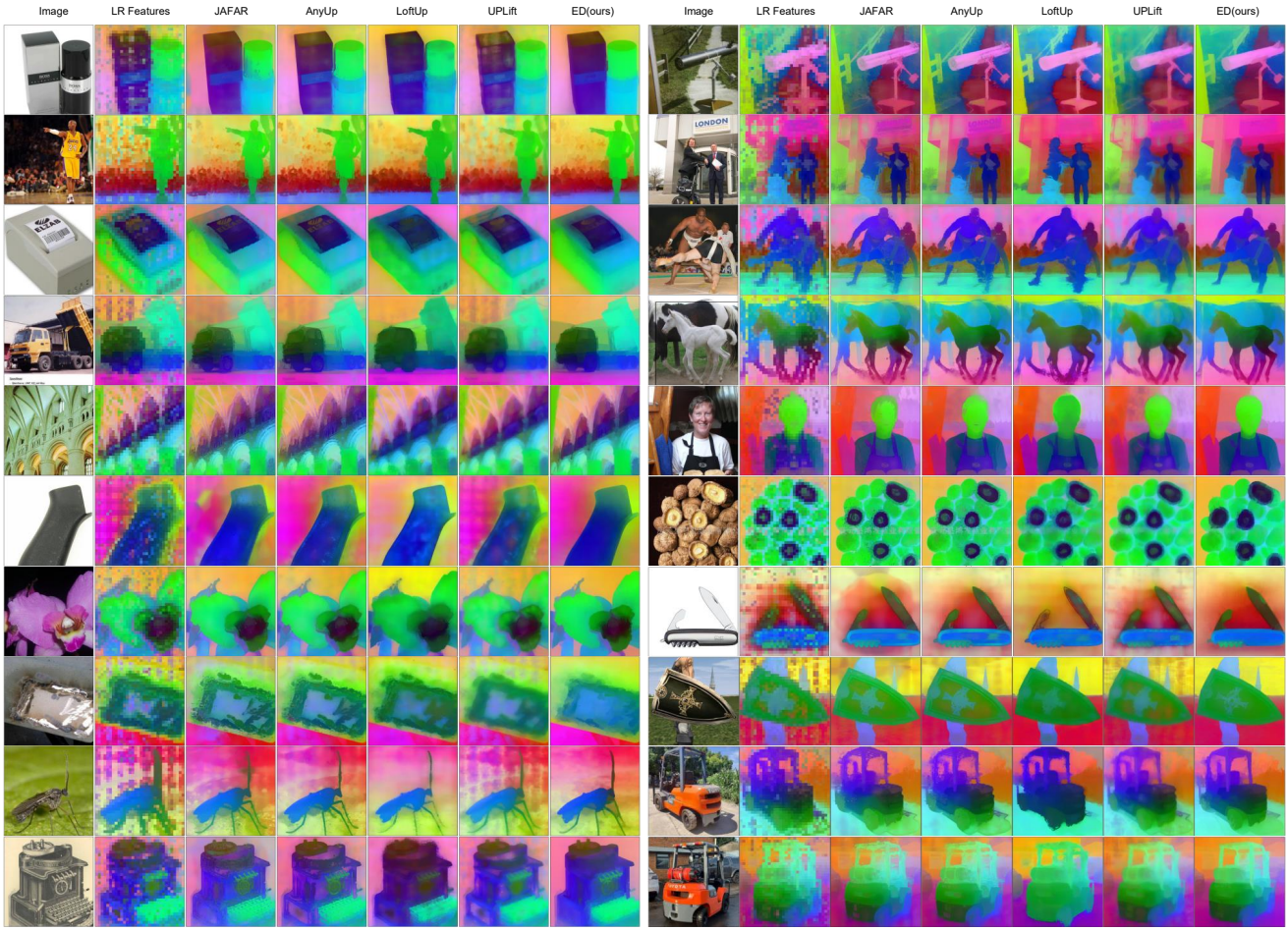


Figure 6. Additional qualitative comparisons through visualizations of the first three PCA components of the resulting features. From left to right: Input image, Original DINOv2 low-res features, JAFAR [5], Anyup [16], LoftUp [7], UPLift [15], ours. Zoomed-in viewing recommended.

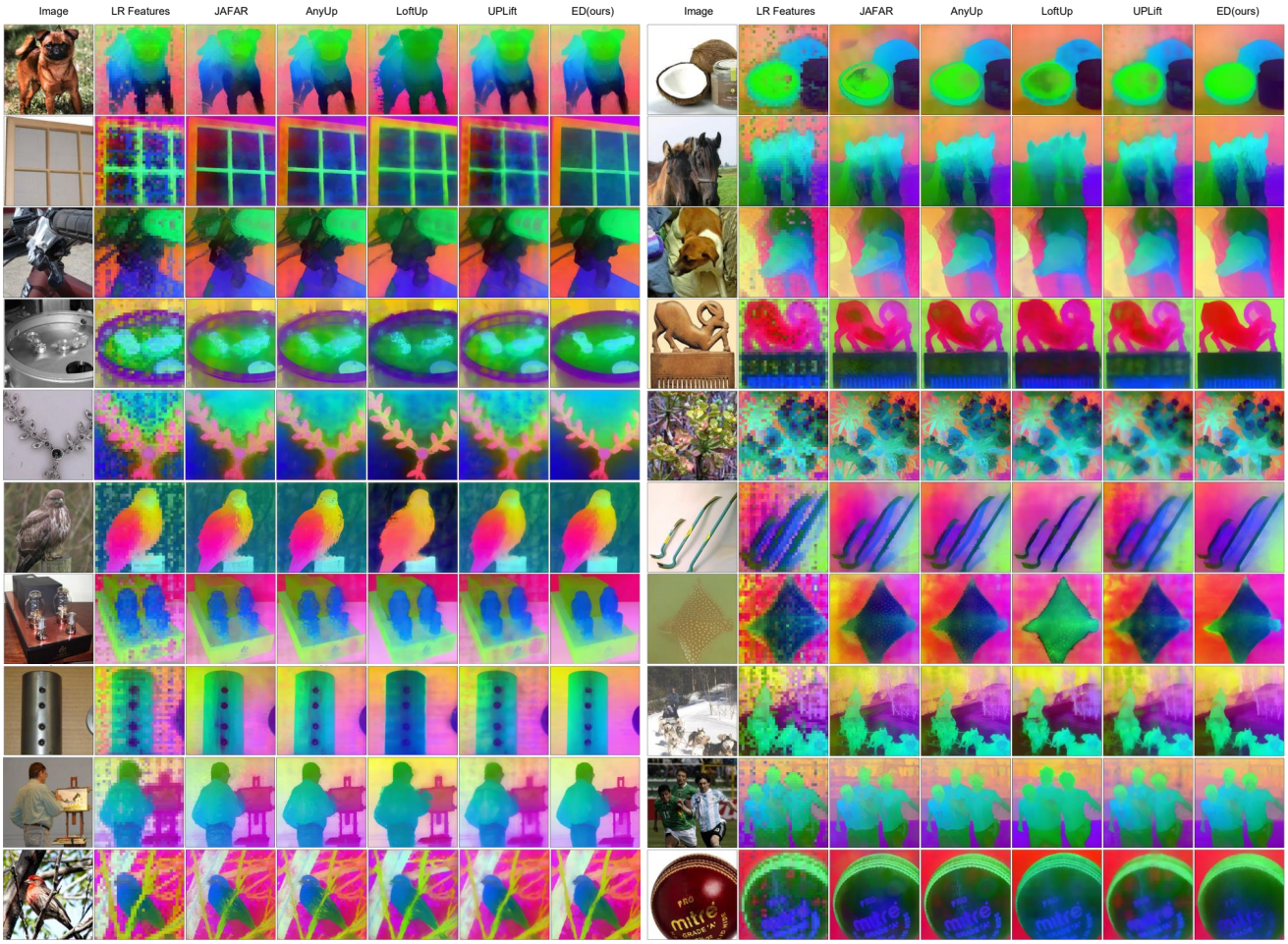


Figure 7. Additional qualitative comparisons through visualizations of the first three PCA components of the resulting features. From left to right: Input image, Original DINOv2 low-res features, JAFAR [5], Anyup [16], LoftUp [7], UPLift [15], ours. Zoomed-in viewing recommended.

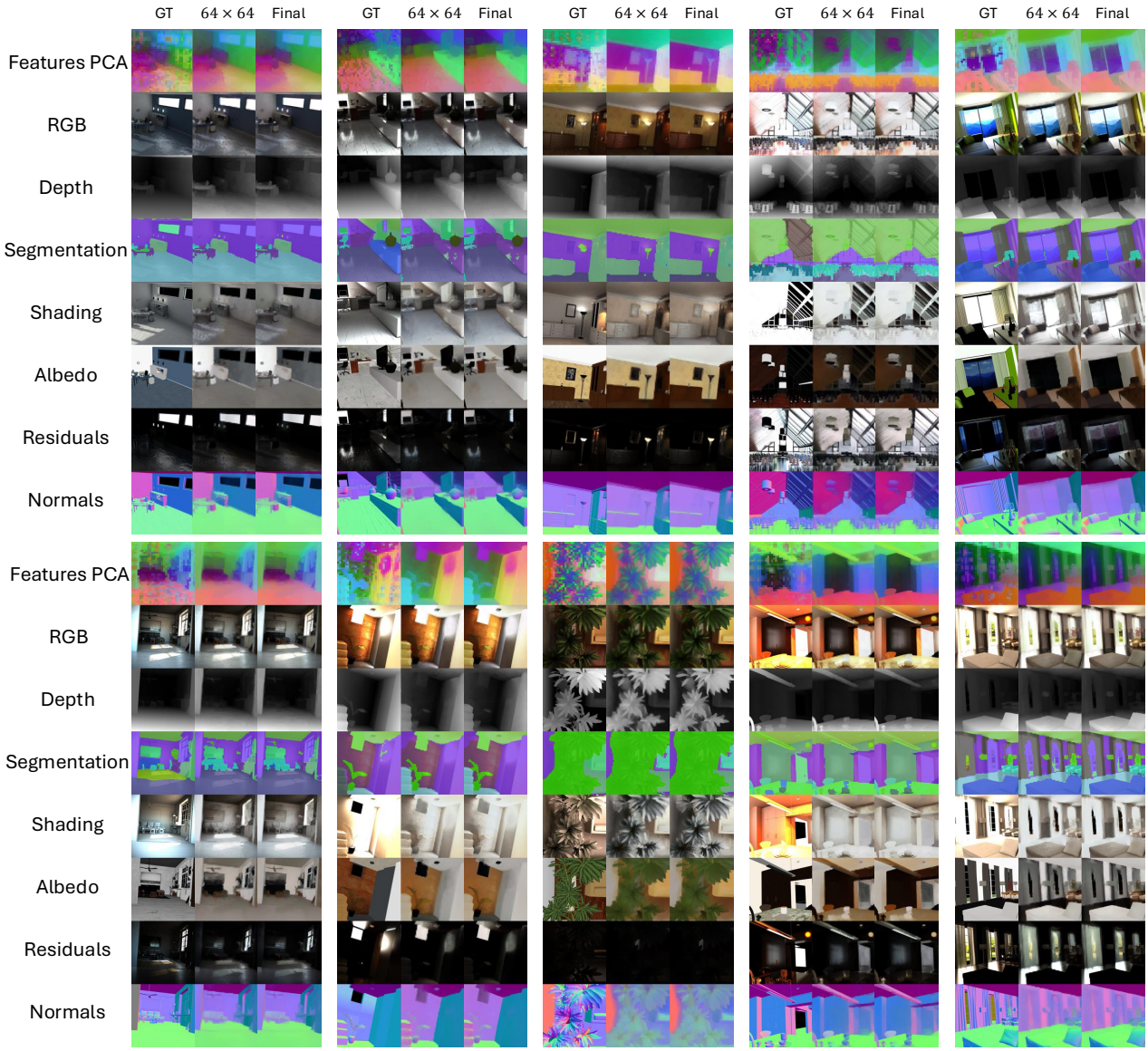


Figure 8. Qualitative evaluation of our auxiliary training tasks. During training, our encoder is trained various tasks, each with an individual head and the minimal receptive field of one pixel. Their results are demonstrated here. Top to bottom: PCA of the upscaled features, RGB reconstruction, depth estimation, semantic segmentation, and four tasks of intrinsic decomposition (shading, albeo, residuals, normals). For each image, we show (left) its GT labels (or the features before upscaling for the PCA), intermediate results of our heads after upscaling to size 64×64 (middle), and the final results of size 256×256 (right). Zoomed-in viewing recommended.

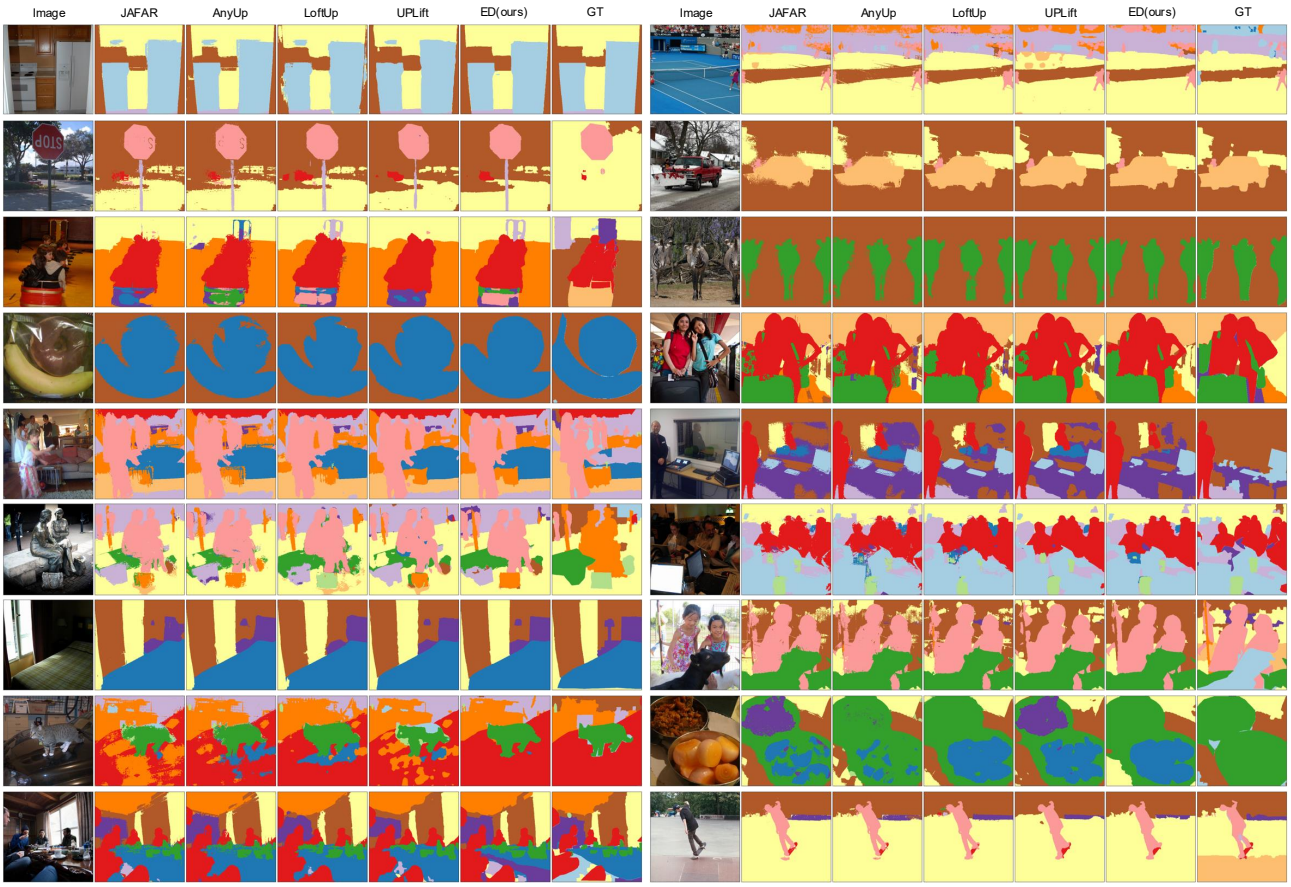


Figure 9. Additional qualitative comparisons for the segmentation task over the COCO dataset. From left to right: Input image, JAFAR [5], Anyup [16], LoftUp [7], UPLift [15], ours, Ground Truth. Zoomed-in viewing recommended.

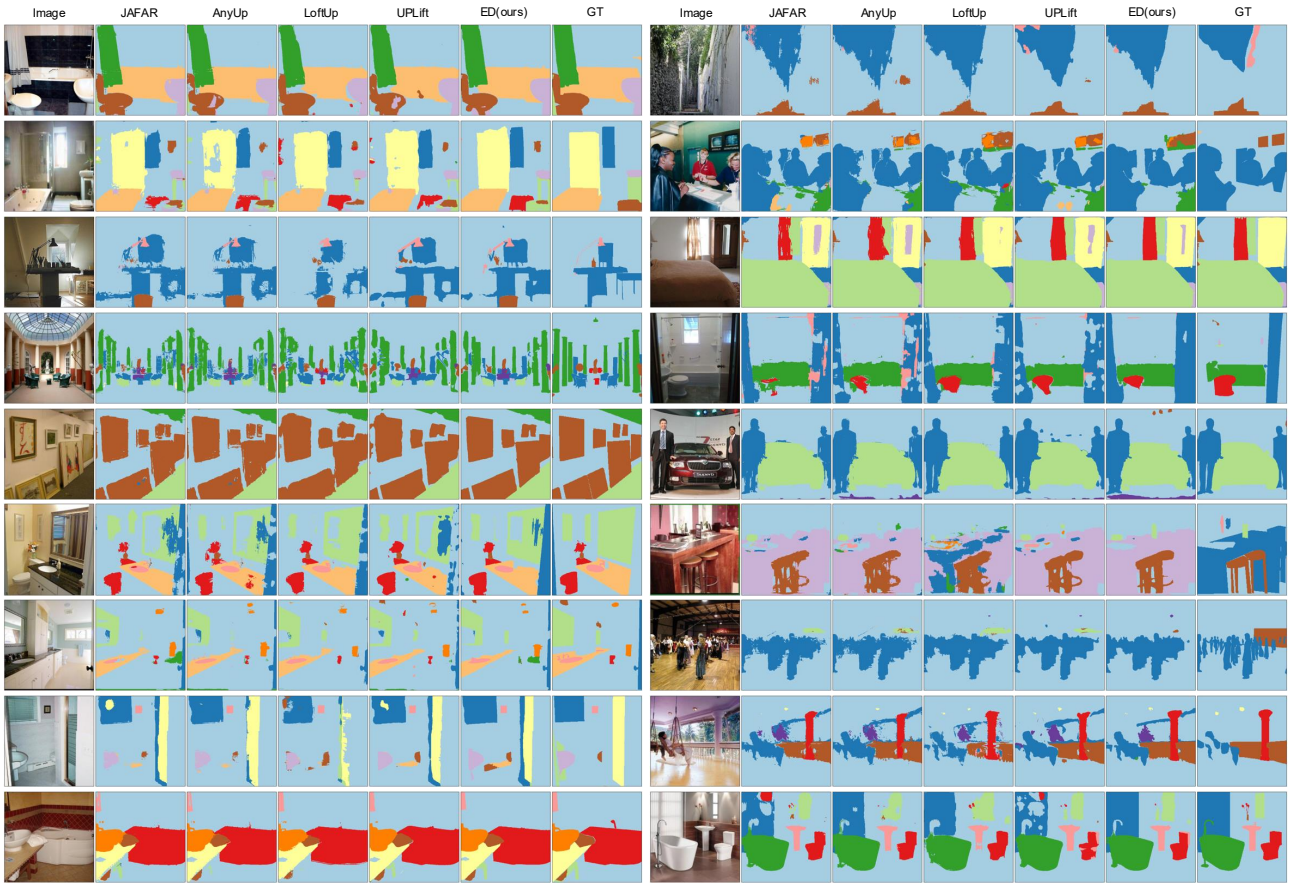


Figure 10. Additional qualitative comparisons for the segmentation task over the ADE20K dataset. From left to right: Input image, JAFAR [5], Anyup [16], LoftUp [7], UPLift [15], ours, Ground Truth. Zoomed-in viewing recommended.



Figure 11. Additional qualitative comparisons for the segmentation task over the VOC dataset. From left to right: Input image, JAFAR [5], Anyup [16], LoftUp [7], UPLift [15], ours, Ground Truth. Zoomed-in viewing recommended.

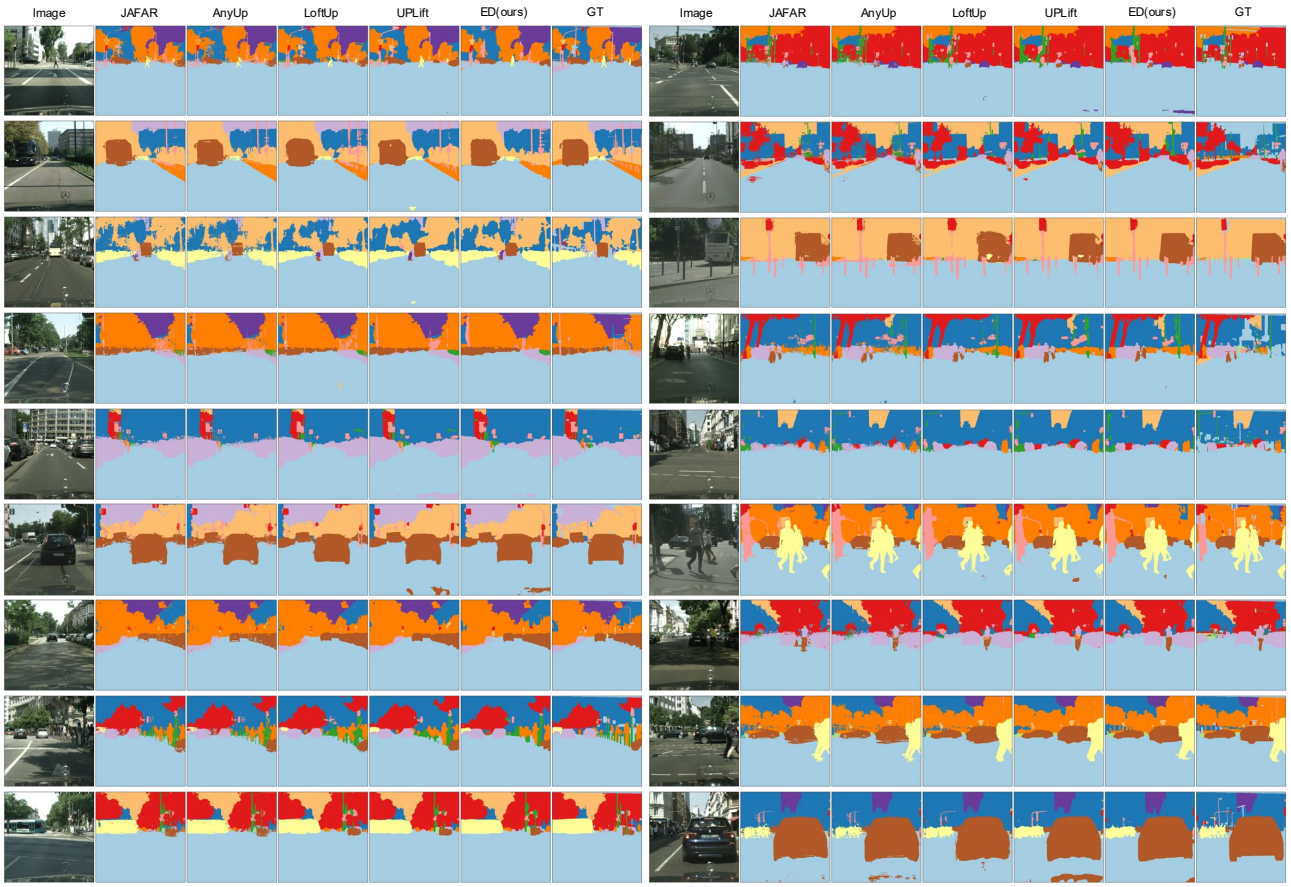


Figure 12. Additional qualitative comparisons for the segmentation task over the CityScapes dataset. From left to right: Input image, JAFAR [5], Anyup [16], LoftUp [7], UPLift [15], ours, Ground Truth. Zoomed-in viewing recommended.

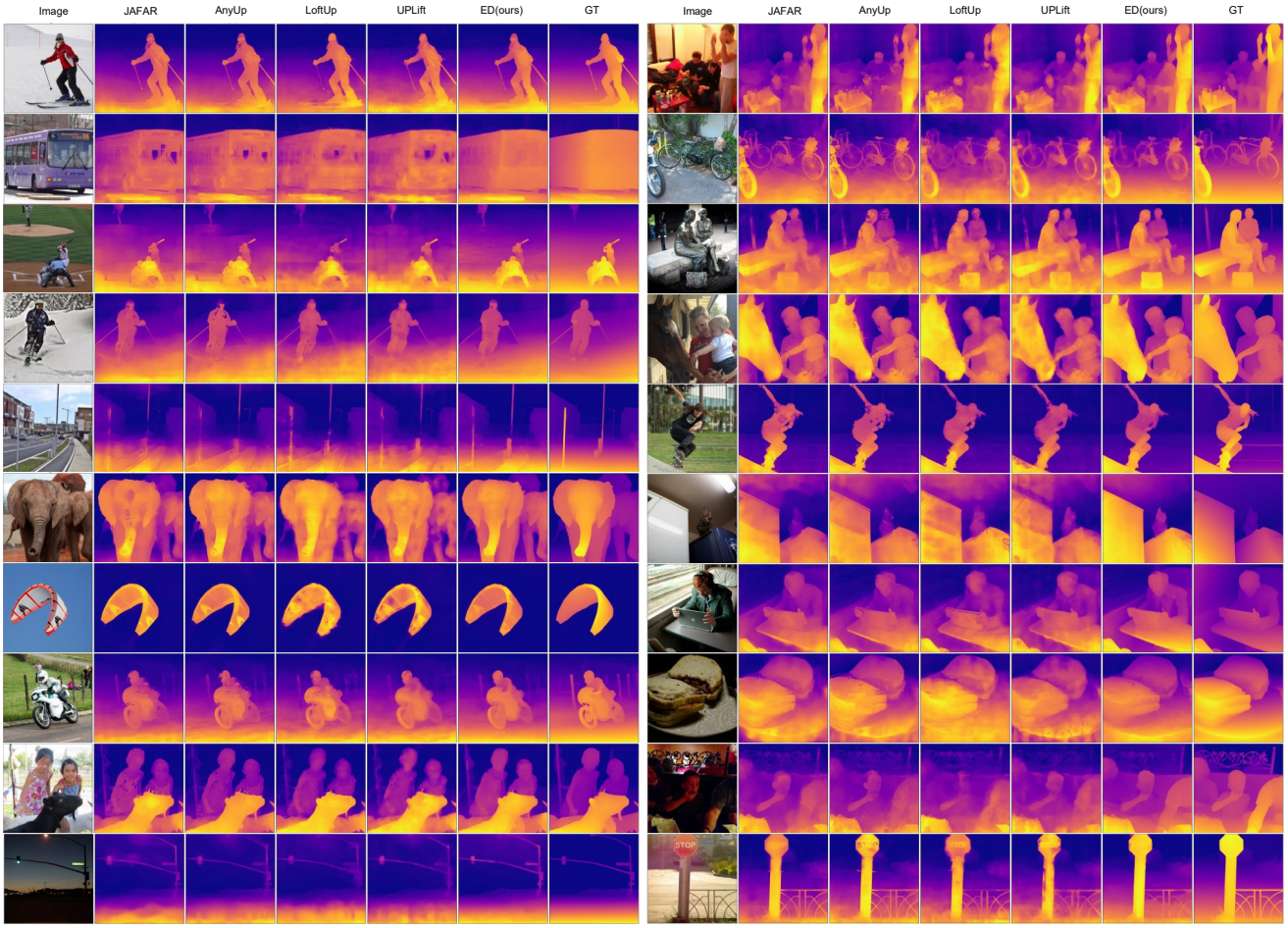


Figure 13. Additional qualitative comparisons for the depth estimation task over the COCO dataset. From left to right: Input image, JAFAR [5], Anyup [16], LoftUp [7], UPLift [15], ours, Ground Truth. Zoomed-in viewing recommended.

References

[1] Simon Baker, Daniel Scharstein, James P Lewis, Stefan Roth, Michael J Black, and Richard Szeliski. A database and evaluation methodology for optical flow. International journal of computer vision, 92(1):1–31, 2011. 2, 5, 6

[2] Bowen Cheng, Ishan Misra, Alexander G. Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. arXiv, 2021. 1, 2, 3

[3] François Chollet. Xception: Deep learning with depthwise separable convolutions, 2017. 1

[4] Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. The cityscapes dataset for semantic urban scene understanding, 2016. 2, 3

[5] Paul Couairon, Loick Chambon, Louis Serrano, Jean-Emmanuel Haugeard, Matthieu Cord, and Nicolas Thome. Jafar: Jack up any feature at any resolution, 2025. 1, 2, 3, 7, 8, 10, 11, 12, 13, 14

[6] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In 2009 IEEE conference on computer vision and pattern recognition, pages 248–255. Ieee, 2009. 1

[7] Haiwen Huang, Anpei Chen, Volodymyr Havrylov, Andreas Geiger, and Dan Zhang. Loftup: Learning a coordinate-based feature upsampler for vision foundation models. arXiv preprint arXiv:2504.14032, 2025. 1, 2, 7, 8, 10, 11, 12, 13, 14

[8] Jitesh Jain, Jiachen Li, MangTik Chiu, Ali Hassani, Nikita Orlov, and Humphrey Shi. Oneformer: One transformer to rule universal image segmentation, 2022. 2, 3

[9] Alexandre Lacoste, Alexandra Luccioni, Victor Schmidt, and Thomas Dandres. Quantifying the carbon emissions of machine learning. arXiv preprint arXiv:1910.09700, 2019. 1

[10] Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, and Piotr Dollár. Microsoft coco: Common objects in context, 2015. 2, 3

[11] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization, 2019. 1

[12] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. arXiv preprint arXiv:2304.07193, 2023. 1, 2

[13] Mike Roberts, Jason Ramapuram, Anurag Ranjan, Atulit Kumar, Miguel Angel Bautista, Nathan Paczan, Russ Webb, and Joshua M. Susskind. Hypersim: A photorealistic synthetic dataset for holistic indoor scene understanding, 2021. 1, 2, 4

[14] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3, 2025. 1

[15] Matthew Walmer, Saksham Suri, Anirud Aggarwal, and Abhinav Shrivastava. Uplift: Efficient pixel-dense feature upsampling with local attenders. arXiv preprint arXiv:2601.17950, 2026. 1, 2, 7, 8, 10, 11, 12, 13, 14

[16] Thomas Wimmer, Prune Truong, Marie-Julie Rakotosaona, Michael Oechsle, Federico Tombari, Bernt Schiele, and Jan Eric Lenssen. Anyup: Universal feature upsampling. arXiv preprint arXiv:2510.12764, 2025. 1, 2, 7, 8, 10, 11, 12, 13, 14

[17] Lihe Yang, Bingyi Kang, Zilong Huang, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything: Unleashing the power of large-scale unlabeled data, 2024. 1

[18] Chong Zhou, Chen Change Loy, and Bo Dai. Extract free dense labels from clip. In European Conference on Computer Vision (ECCV), 2022. 1